Remember the evidence. Choose where to focus. Bound the work.

Omega Hive × ASI

Private memory, independent attention and guarded native inference

THROUGH M3 · C1–C4

REAL HIVE · DETTA · PLN

Architecture and evidence • 16 October 2026 • supported local scope

ELI16: remember, weigh evidence, do the calculation

Think of a student investigating a question with a notebook, evidence ledger and calculator.

C1 · NOTEBOOK

Keep the original story

Record what happened and where each report came from. Keep contrary observations; replay does not add another event.

C2 · EVIDENCE LEDGER

Know what supports a belief

A copy is still the same source. A conclusion keeps its parents; it does not become a fresh observation.

PLN · CALCULATOR

Apply a guarded formula

Use the existing pinned mathematics on exact inputs. C2 checks and saves the result; the calculator owns no beliefs.

What this gives us

An inspectable trail from original observations to admitted conclusions.

A repeated message, a new conclusion and an independent observation mean different things.

ELI16: a priority list and a limited attempt budget

There are two prepared questions, but time for only one calculation. Which should run first?

C3 · Priority list

Give each eligible task an integer score. Rank it, decay it explicitly, and remember feedback once.

C4 · Work coordinator

Choose from that list. Save the attempt reservation before calculation, then track admission and feedback.

Restart · Same budget

Reopening the notebook does not refill the budget. Recover a saved decision; do not credit it twice.

Observed in real Hive

One native action, one admitted conclusion, one feedback credit; restart and replay tested.

The deliberate tradeoff

A crash can spend the last attempt without producing a conclusion. The receipt says so.

Priority is not truth. A bounded step finishing is not a promise that a conclusion was found.

Fifteen modules; four responsibility groups

A module owns a responsibility. It need not be a separate process or repository.

CLIENT / COGNITION

- Episodic memory
- Belief revision
- PLN · one rule
- STI attention
- Inference control

APPLICATION PORTS

- AtomStorage
- Relations · future
- Search projection · future
- Agent messaging
- Delegated execution

CONTROL & POLICY

- Registry
- Compatibility resolver
- Authorization

Local test policy
and static descriptors

PROVIDER ADAPTERS

DA storage

Real local DeTTa

Consensus / finality

M1 historical; M3 NOT_RUN

SPECIFIED / FUTURE IMPLEMENTED LOCAL REAL PROVIDER

M3: one grounded rule; no ECAN or LTI

Data responsibilities: bytes → references → history

Persistence, identity, structure, discovery and provenance have different owners.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
DA storage	Persist opaque bytes put · get · availability	Payloads + shares REAL LOCAL PROVIDER
AtomStorage	Scoped immutable envelopes create · get · getMany · availability	Reference + replay metadata IMPLEMENTED
Relations	Represent ordered structural roles assertRelation · getRelation	Immutable relation records SPECIFIED
Search projection	Find refs at a known watermark query	Rebuildable index SPECIFIED
Episodic memory C1	Record / append / read ordered events pendingEvidence · acknowledgeEvidence	Private history + delivery intents IMPLEMENTED · M2

DA persists bytes. Only the application owners interpret events, claims or beliefs.

Cognitive responsibilities: four separate roles

M3 adds a pure grounded kernel, independent STI and a bounded coordinator.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / BOUNDARY
Belief revision C2	submitEvidence · readBelief submitDerivation · readDerivation	Supports, snapshots, attempts Only C2 commits belief changes
PLN kernel	Apply one grounded rule to exact premises profiles · applyRule	No mutable domain state Pure native proposal
Attention C3	submitStimulus · select · advance	STI, clock, causal receipts No cognition dependency
Inference control C4	Run one immutable workset step	Selection, budgets, progress No truth or attention ownership

C4 calls C2 for admission. C2 invokes the kernel, then checks authority and commits its decision.

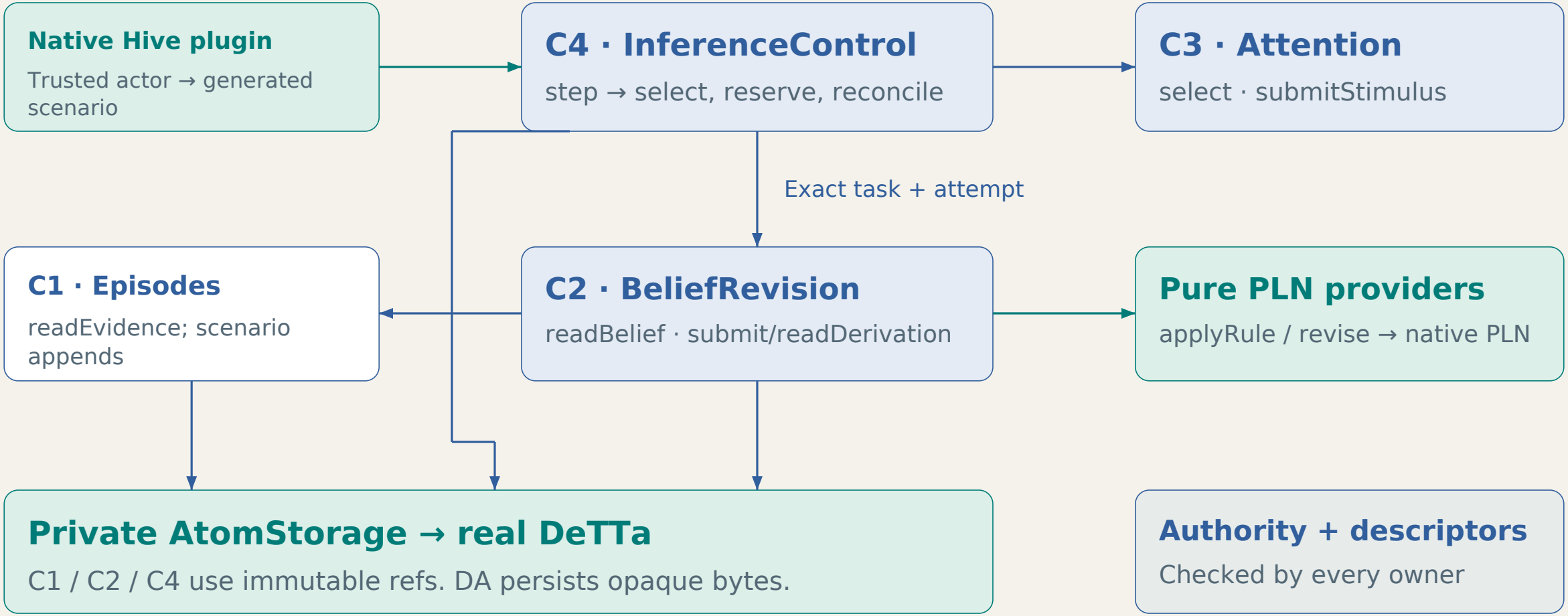
Control responsibilities: move work, preserve authority

The remaining six modules connect owners and report specific facts.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
Messaging	Accept, deliver, acknowledge send · poll · ack	Inbox + deduplication · IMPLEMENTED
Execution	Run one named operation submit · status · result	Jobs + completion outbox · IMPLEMENTED
Registry	Describe configured providers descriptors	Static configuration · IMPLEMENTED
Resolver	Match required semantics resolve	No mutable state · IMPLEMENTED
Consensus / finality	Submit approved commitment; observe submit · observe	Signed intents · HISTORICAL M1
Authorization	Check current caller + scope + method authorize / live policy checks	Operator policy · LOCAL FIXTURE

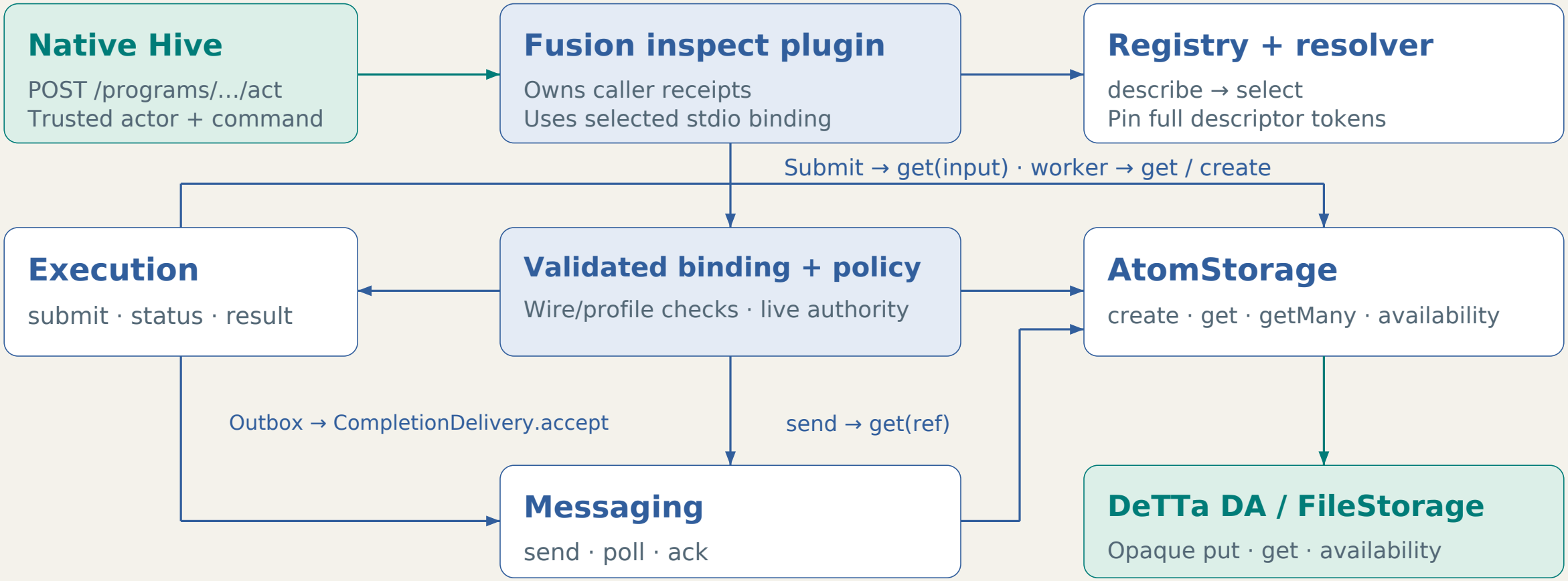
Function dependencies: the exercised M3 path

Solid arrows point from caller to provider. Owners expose ports, never each other’s tables.



Function dependencies: the preserved M1.2 path

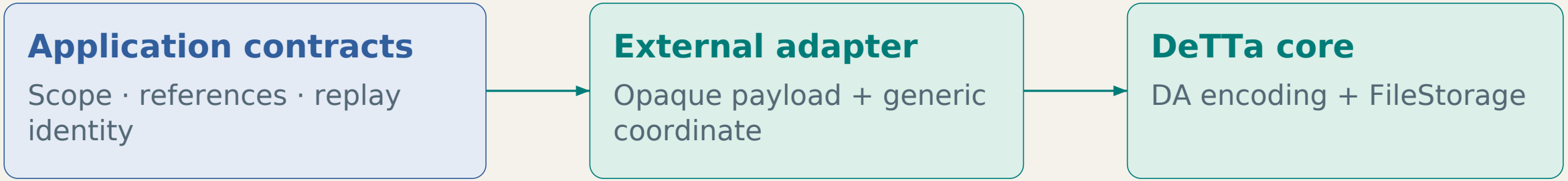
Solid arrows = exercised calls. Every provider effect crosses an existing validated interface.



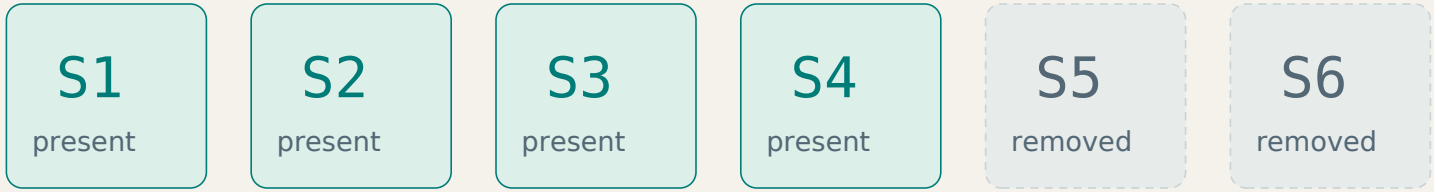
Optional finality uses a separate caller + approved digest + signer; M1.2 does not invoke it.

The DA boundary remains application-neutral

The external adapter translates application references into generic coordinates and opaque bytes.



4 data shares + 2 parity shares



Observed reconstruction

Cached payload + two shares removed: exact bytes recovered. A third missing share is unavailable.

The DA core does not interpret Hive identities, truth values, attention balances or job state.

Two independent substitutions, one design principle

Select semantics and capabilities explicitly; keep callers independent of provider internals.

AtomStorage · preserved M1/M2

Real local DeTTa or an independent atomic-file fixture.

In-process and private stdio bindings pass the same workflow.

Attention · new in M3

SQLite owner or an independent atomic-file fixture.

Same STI policy, tasks and budgets; unchanged consumer produces matching outcomes.

No shared owner transitions, inheritance or private-table access in the attention replacement.

These are fresh-store replacements. Existing accepted state is not migrated.

Private state has explicit owners and placement

A trusted Hive actor maps to an operator-configured private scope and credential.

PRIVATE LOCAL COMPOSITION

C1 · Episodes

History + delivery intents

C2 · Beliefs

Supports + exact attempts

C3 · Attention

STI + clock + causal replay

C4 · Controller

Budgets + progress + feedback

Public Hive receipt

Generated counters, status and public command receipt ID.

Private trace

Task/support hashes, premises, lineage, attention and payloads stay private.

Shared M1 wire and DA contracts stay unchanged; every owner rechecks current authorization.

Executable contracts make the inference step finite

The supported task is explicit, immutable and grounded before the controller starts.

CONTRACT	WHAT IS FIXED	WHY IT MATTERS
Context + task	Private context; exact role references	Labels cannot merge distinct identities
Workset	1–16 unique immutable tasks	No search or automatic task expansion
Premise snapshot	Claim / context / revision / support ID	No silent substitution with latest state
Derived support	Closed tag + inherited roots / ancestry	No fresh observation or evidence count
Work budget	Candidates, attempts, generation, deadline	Restart cannot silently refill the allowance

Profile mismatch, malformed response or changed semantic intent fails explicitly.

C1: one occurrence, one position in the story

Immutable event atoms sit behind a stable episode identity and a committed sequence.

EpisodeRef = scope + ID

1 · Report

No numerical claim

2 · Assessment

Explicit strength / confidence

3 · Copy

Inherits original roots

4 · Contrary

Kept in the history

Same occurrence + same body

Return the original receipt, even under a new request key. The head does not advance.

Same occurrence + changed meaning

Conflict on body or provenance changes. Distinct real occurrences may share identical bytes.

C1 commits event + evidence intent together; AtomStorage has already committed independently.

C2: admitting evidence and changing belief are separate

New observations remain visible even when they cannot change the current aggregate.

Evidence submission

C1 evidence reference or an existing C2 support; exact claim and context.

Admission decision

ADMITTED or DUPLICATE
Track occurrence and provenance.

Numerical decision

Initialize, revise, no change, retain alternative, or model-inapplicable.

Aggregate with exact coverage

Every numerical result names the contributing leaves, origins, premises, formula and assumption.

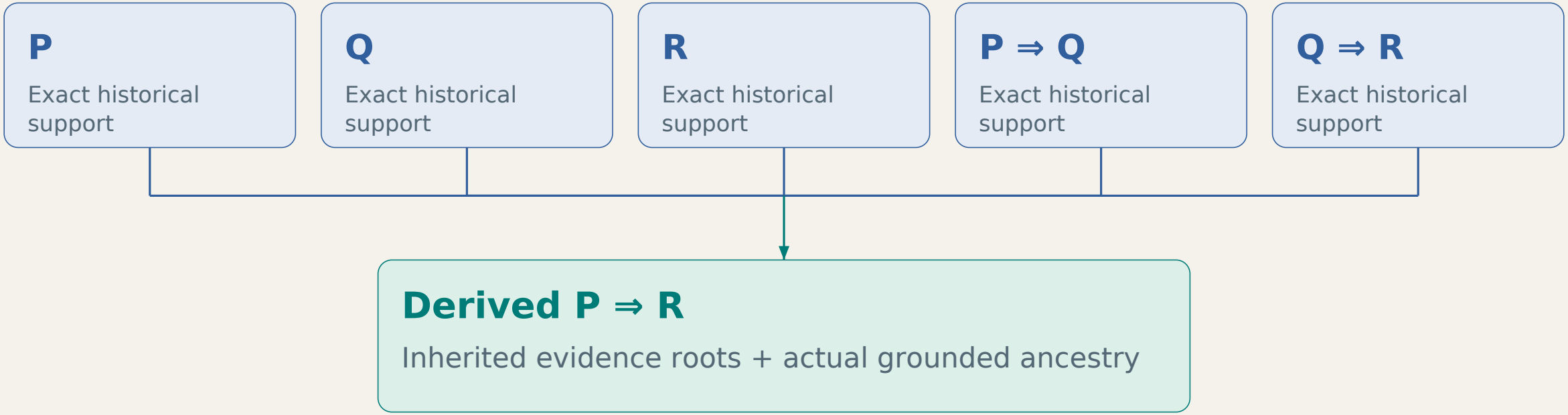
Unaggregated alternatives

Copied or contrary observations remain in the snapshot. Later eligible assumptions can combine existing supports.

One saved truth value is not a summary of every observation. Read its coverage and alternatives.

A derived conclusion inherits its evidence trail

The result is a new support record, while its original observations and ancestry remain the same.



Unknown target: initialize inferential support. Existing truth: keep the alternative.

No new observation, root, independence assertion or evidence count is created.

One grounded rule reuses the actual native PLN

No replacement arithmetic and no general search loop: five explicit premises produce one proposal.

Fusion PLNKernel

Exact bindings, source/profile checks and remaining deadline

→

Pinned Reachability

PLNAdapter.apply_rule
DeductionRule + guards

→

Native PLN / PeTTa

Truth_Deduction
SWI 10.1.14 + Janus

Two probability checks

Upstream exact Python precheck and the native consistency guard both apply.

Separate compatibility profile

Deduction has its own source/runtime allowlist; the revision-only profile stays intact.

Kernel output is a proposal. C2 performs final authorization, CAS and admission.

Native goldens check deduction and preserved revision

Independent expected values check pinned native formulas; no replacement math runs in the owners.

M3 · Grounded deduction

P, Q, R = (0.5, 0.8)
P⇒Q, Q⇒R = (0.8, 0.7)

Expected P⇒R: (0.68, 0.3136)

M2 · Guarded revision

(0.75, 0.50) + (0.25, 0.50)
→ (0.50, 2/3)

Unequal confidence 0.50 / 0.75
→ (0.375, 0.80)

`Observed deduction: 0.6800000000000002 / 0.31360000000000005`

Absolute tolerance 1e−12; finite-domain and lineage guards remain active.

These are model-specific heuristics, not calibrated real-world confidence or proven entailment.

C3: a small, independent attention policy

sti-additive-linear-decay/1 · integer STI · explicit logical time · persistent causal identity

Stimulate

Add within per-target saturation. Refuse the whole change if the context budget would be exceeded.

Select

Rank positive eligible targets by descending STI, then the full canonical reference. Reads do not decay state.

Advance

Explicitly decay by logical steps. Stored causal generation survives decay, new ingress and new step IDs.

Default limits: 100 STI per target · 300 per context · decay 1 per logical step

Only authority and opaque-reference syntax are required; no Hive, storage or PLN runtime.

This reference policy is implemented. ECAN, LTI, diffusion and new attention research are out of scope.

Attention replacement preserves the measured outcome

Same tasks, evidence, policy, budgets and deadline; only provider and declared preference vary.

ATTENTION PROVIDER	PREFER	SELECTION ORDER	STEP TIME
SQLite	B	b → a	0.744361 s
SQLite	A	a → b	0.677157 s
Atomic-file fixture	B	b → a	0.726126 s
Atomic-file fixture	A	a → b	0.641150 s

Every case

1 native call · 1 unique admission · 1 feedback credit

Reopen all owners

Same receipt · zero additional native calls

Four local observations demonstrate substitution and selection effect; they are not a performance ranking.

C4: reserve work durably before invoking the rule

A work budget limits physical evaluation opportunities, even when the process dies.

1 · Freeze selection

Read the immutable workset and exact attention order.

2 · Reserve attempt

Commit candidate progress and a physical attempt slot.

3 · Ask C2

Recover a decision or dispatch the exact derivation attempt.

4 · Reconcile

Save decision and causal feedback acknowledgment.

Budget survives restart

Candidate count, reserved attempts, generation limit and absolute deadline are persisted.

Selection changes next step

Current selection is frozen. A unique admission receives one unit of causal feedback.

No refund for a lost pending attempt. A new evaluation needs another slot and attempt identity.

Five crash boundaries, independent owner commits

10 direct boundary cases + 5 actual Hive caller boundary cases in the recorded M3 run.

PROCESS DIES AFTER…	DURABLE FACT	RECOVERY RESULT
Attention stimulus commit	C3 causal receipt	Replay does not add STI again
C4 attempt reservation	Slot already spent	One-slot budget can end without a result
Kernel return; before C2 commit	Pending attempt; no admission	Another evaluation needs another slot
C2 belief commit	Authoritative saved decision	Recover without another native call
C3 feedback commit; before C4 ack	Credit recorded by logical cause	Recover ack without a second credit

The evidence proves tested local process recovery, not a distributed transaction or power-loss durability.

Authorization, replay and profiles remain live

Cached success does not bypass current authority, and compatibility is never silently widened.

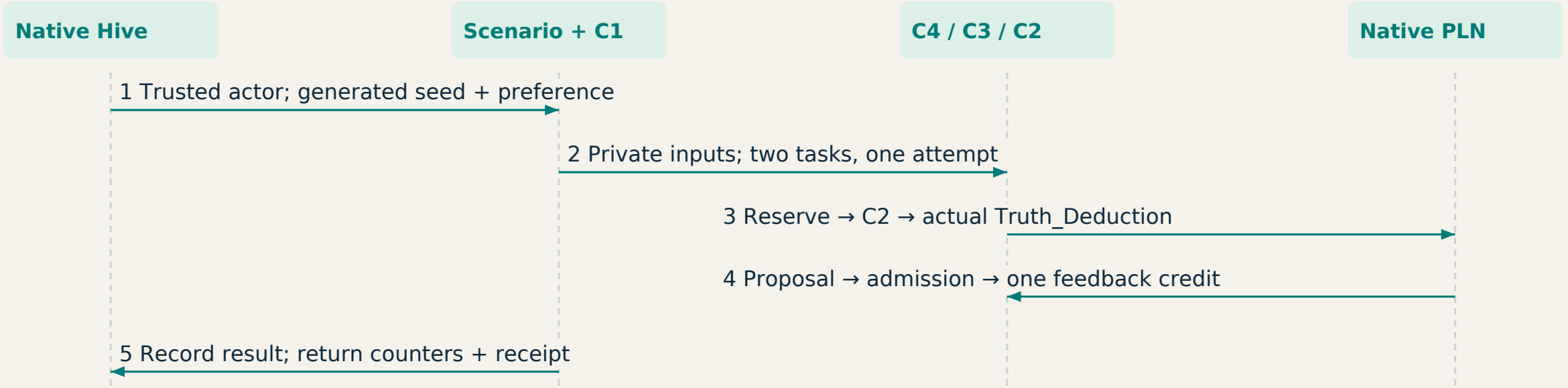
TRIGGER	OBSERVED DECISION	PRESERVED BOUNDARY
Same key, altered nested request	CONFLICT / validation refusal	Exact immutable semantic intent
Permission revoked before replay	DENIED	Live authority even for cached receipts
Authority / CAS changes during math	Refuse final admission	C2 rechecks before its own commit
Runtime or response binding changes	Explicit compatibility failure	No weaker fallback or invented result
Clock advances before feedback	Pending CONFLICT	No implicit retiming of causal feedback

Attention ingress and feedback have distinct trusted bindings; selection never grants authority.

One real Hive action reaches the entire private path

Existing authenticated Programs runtime → fusion-inference / infer-generated.

```
POST /api/programs/fusion-inference/act
```



One private episode: 10 generated observations + action + result. Public receipt contains no private trace.

The same one-attempt budget makes attention visible

Two eligible tasks have exact premises. Initial STI selects which one can consume the attempt.

Prefer task A

A	20
B	10

First attempt → A

Prefer task B

A	10
B	20

First attempt → B

Both attention providers preserve this effect; replay adds no native call and no second credit.

Attention changes scheduling. It does not alter premise truth values or manufacture independent evidence.

Real providers and fixtures remain visibly separate

A real runtime can execute generated inputs. Each claim carries its own evidence label.

CLASSIFICATION	EXERCISED COMPONENTS	WHAT THE RESULT SUPPORTS
REAL PROVIDERS	SQLite STI · local DeTTa · pinned native PLN Actual Hive Programs / Supervisor / PeTTa	Native calls, owner persistence and tested runtime integration
FIXTURES	Generated tasks, observations, authority policy Independent atomic-file attention; guard/fault inputs	Conformance, controlled replacement and deliberate refusal behavior
BLOCKED	None for supported M3 prerequisites Historical exact chain source unavailable	No missing M3 prerequisite is hidden by a fixture fallback
NOT_RUN	Chain, power loss, distributed failover, empty-cache build, production quality/throughput	No claim for these extensions or excluded work

Executed tests cover contracts, native calls and crashes

Counts overlap. The 51 original and 16 M2 fixture subsets are not added again to the 118-case matrix.

COMMAND / LAYER	CASES	EVIDENCE TYPE
Preserved M1/M2 regression	118 matrix	90 original + 16 M2 + 12 M3; native suites also rerun
Attention / contract / upstream guards	5 / 7 / 15	Fixtures; C3 standalone; RecordingRuntime guards
Native kernel + derived admission	13	Real goldens plus explicitly injected faults
Direct real process kills	3 methods / 10 cases	SIGKILL across independent commit boundaries
Actual native Hive	8	Zero failures, errors or skips; five more caller kills
Controlled attention replacement	4	Fresh cases, native PLN + DeTTa, exact replay

Final execution used clean detached 1af3979. Earlier failures and an interrupted run remain separately labelled.

M3 gates: contracts and independent cognition

All 13 required gates passed for the supported local scope; no blocked M3 prerequisites.

GATE	RESULT	RECORDED EVIDENCE
BASELINE	PASS	M1/M2 behavior, sources and selected native tests
CONTRACTS	PASS	Exact immutable tasks, vectors and owner responses
C3	PASS	Persistent bounded STI and causal replay
C3_INDEPENDENT	PASS	No PLN, Hive, C1/C2 or DA runtime dependency
ATTENTION_SWAP	PASS	Independent same-profile replacement; four cases
KERNEL_NATIVE	PASS	Actual guarded deduction; independent golden
DERIVED_ADMISSION	PASS	Historical premises; lineage; no fresh evidence

Fixture, real-provider, blocked and unrun results retain separate meanings.

M3 gates: bounded operation and real integration

All 13 required gates passed for the supported local scope; no blocked M3 prerequisites.

GATE	RESULT	RECORDED EVIDENCE
C4_BOUNDED	PASS	Persistent attempts, candidates, deadline and generation
RECOVERY	PASS	10 direct + 5 native Hive SIGKILL cases
PRIVACY	PASS	Private canaries absent from tested public surfaces
HIVE_M3	PASS	Eight actual authenticated native integration cases
REPRO	PASS	Fresh native objects / venv / DeTTa target
SCOPE	PASS	C3/C4 + required admission extension; one rule
CHAIN	NOT_RUN	Excluded; no chain action or deployment

Fixture, real-provider, blocked and unrun results retain separate meanings.

Reproducibility connects source, build and observed result

Exact implementation and provider identities make the evidence inspectable.

LAYER	RECORDED STATE	BOUNDARY
Requested / actual start	f0cee24 → 926882a	Only presentation/docs and M3 brief intervened
Clean tested implementation	1af39794649d	Separate detached checkout; final verifier exit 0
Evidence snapshot	c4498ce3047f	No implementation/test changes after verified SHA
Fresh provider build	354.696 seconds	New native objects, venv and DeTTa target
Pinned runtime	SWI 10.1.14 · Python 3.12.12 · Janus 1.5.3	Dedicated deduction profile; no upstream patches

Warm download caches are explicit. The fresh binary digest identifies this build, not cross-host bit reproducibility.

The limitations are part of the contract

Durable local progress is useful even when the receipt correctly reports unfinished work.

Conservative crash budget

A lost pending attempt consumes its slot. A completed bounded step need not have a conclusion.

Pending feedback

Concurrent clock advance can leave CONFLICT. Deadline expiry can leave work unacknowledged.

Explicit C1/C2 compatibility

M3 v3 belief snapshots do not fit M2 v2 C1 acknowledgments. C1 delivery intents remain pending.

Local assumptions

Same-host clock, intact files and trusted operator. No hard real-time, power-loss or distributed failover claim.

Stopped at M3: no LTI, general search, new attention research, live migration or chain work.

Keep the evidence.
Separate the decisions.
Make the budget real.

Real Hive + native PLN + replaceable attention
with explicit private owners and tested recovery.

M3 COMPLETE · 13 REQUIRED GATES PASS

Implementation map: responsibilities stay local

Follow interfaces and owner journals before following concrete provider classes.

RESPONSIBILITY	FILES / SURFACE	OWNED STATE
C1 / original C2	episodes.py · beliefs.py · evidence_relay.py	Episode history; revision-only journals
C2 derived admission	derived_beliefs.py · cognition_contracts.py	Fresh derived-beliefs.sqlite · local v3
C3 attention	attention.py · adapters/fixture_attention.py	SQLite or independent atomic-file state
Pure PLN providers	adapters/pln_deduction.py · pln_revision.py	No mutable domain state
C4 controller	inference.py · deadlines.py	inference.sqlite · budgets / progress
Composition / native caller	cognition.py · hive/inference_program/	Explicit provider bindings; caller receipt

Exact sources identify the verified native stack

Full revisions are printed here; source files and built binary hashes remain in the committed evidence.

Fusion · tested	`1af39794649d0ca95cd729c64868202368da1ec4`
Hive schematics	`4af00db4e395375484dd48d0b9068a59afcf01f8`
Hive	`34b0fb32ab3806e5609fc89bb7535518a35e1aea`
DeTTa	`272888f264dd2dde18b574cabb53c266b0dac2ec`
Reachability	`dd56f3f053ed90239de5377021a7d3b3fbf424ac`
PLN	`4405956947c4b53c7ff01bd565aa3b114bc970a1`
PeTTa	`ae66fa8e41dcd5539d614706bd4e5cfb34f9608d`
SWI 10.1.14	`69775434c8226897626b226aefcc8266499f1e2e`

metta-attention inspected at 3e1c8147; not used as the bounded STI provider.

Profiles and bounds make compatibility explicit

Existing profiles remain selectable; fresh M3 owners opt into new semantics.

PORT / PROFILE	VERSION AND POLICY	KEY BOUND
M1 shared interfaces	Wire 1.0.0 · immutable-bytes/1	Existing storage / messaging / execution preserved
C1 / original C2	Local 2.0.0 · original M2 profiles	Original relay / acknowledgment DTOs
C2 derived	Local 3.0.0 · pln-derived-private/1	Exact history; tagged supports; fresh journal
C3 attention	Local 2.0.0 · additive / linear decay	100 per target; 300 per context; generation 3
PLN kernel	grounded-five-premise-deduction/1	One rule; five exact premises
C4 controller	bounded-grounded-inference/1	1–16 tasks; candidate / attempt / time limits

C1 acknowledgment compatibility remains an open seam

A durable M3 admission and an M2 evidence-delivery acknowledgment are different results.

C1 · Local v2

Keeps the generated episode and its available pending delivery intents.

C2 · Local v3

Direct derivation admission and exact logical replay are durable.

C4 receipt

Records bounded progress, pending feedback and explicit terminal reason.

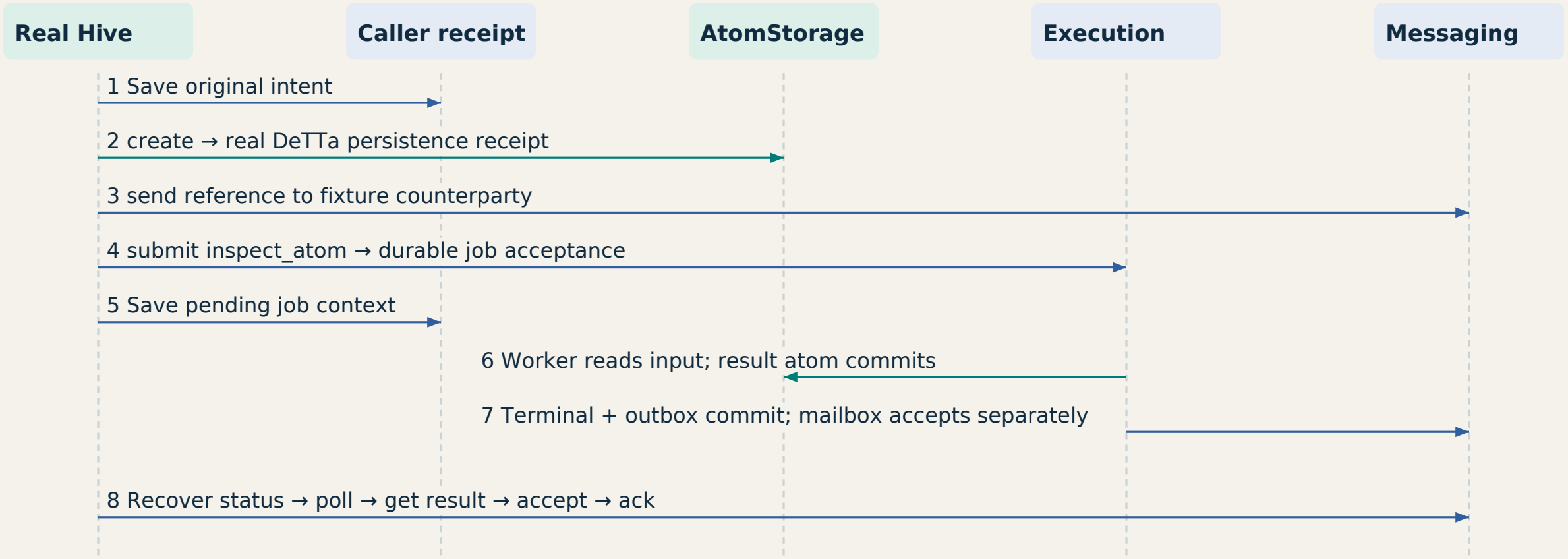
v3 snapshot ≠ v2 acknowledgment DTO

Original M2 relay + acknowledgment tests still pass under the original profile.

No fabricated acknowledgment, silent profile widening or claim that every delivery intent is drained.

One native operation crosses the existing interfaces

The caller stores intent; each provider commits only its own state.



Historical M1: one result reached an isolated chain

This is the earlier protocol-peer artifact. No M1.2, M2 or M3 Hive result was submitted.

PENDING
Signed deploy accepted

→

INCLUDED
Block observed; effects pending

→

FINALIZED
Deploy effects reported finalized

M1 recorded observation

Block height	**3**
Execution error	**false**
Rejection count	**0**
Execution cost	**351 phlo**

Historical commitment digest

```
f9236451ff785dc58315251cf864a3d1
0938412551cf45f448055061bc953ea1
```

One local validator; direct node observations.
Not a consensus proof or production test.

M3 CHAIN = NOT_RUN. Historical source reproduction requires exact F1R3 changes or a separate rebaseline.

Reproduce the supported M3 gates

Use the tested revision or a reviewed descendant, with source access and a fresh destination.

BUILD FRESH PINNED PROVIDERS

```
uv run --python 3.11 --locked python \
scripts/build_m3.py artifacts/m3-providers
```

VERIFY M1/M2 + C3/C4 + NATIVE HIVE + CONTROLLED SWAP

```
uv run --python 3.11 --locked python \
scripts/verify_m3.py artifacts/m3-providers/environment.json
```

Read: manifest.json → final/verification.json → suite logs → final/comparison.json

Source access, fixture tests, native execution and archived evidence are separate reproducibility requirements.

Retry identities answer different questions

Preserve the logical cause while distinguishing transport replay from physical evaluation attempts.

IDENTITY / STATE	MEANING	RECOVERY ACTION
Request / step ID	Exact immutable command intent	Same body replays; changed body conflicts
Logical derivation ID	The same exact grounded conclusion	Deduplicate admission and causal feedback
Attempt key + reserved slot	One physical evaluation opportunity	Pending death spends slot; new key needs budget
Committed C2 decision	Saved authoritative admission outcome	Read first; do not recalculate
C3 causal feedback receipt	One workflow credit for logical admission	Recover acknowledgment; do not credit twice
Deadline / generation watermark	Durable limits across process restarts	Do not reset, refund or implicitly retime